Kinetics of tracheid development explain conifer tree-ring structure

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Summary

- Conifer tree rings are generally composed of large, thin-walled cells of light earlywood followed by narrow, thick-walled cells of dense latewood. Yet, how wood formation processes and the associated kinetics create this typical pattern remains poorly understood.
- We monitored tree-ring formation weekly over 3 yr in 45 trees of three conifer species in France. Data were used to model cell development kinetics, and to attribute the relative importance of the duration and rate of cell enlargement and cell wall deposition on tree-ring structure.
- Cell enlargement duration contributed to 75% of changes in cell diameter along the tree rings. Remarkably, the amount of wall material per cell was quite constant along the rings. Consequently, and in contrast with widespread belief, changes in cell wall thickness were not principally attributed to the duration and rate of wall deposition (33%), but rather to the changes in cell size (67%). Cell enlargement duration, as the main driver of cell size and wall thickness, contributed to 56% of wood density variation along the rings.
- This mechanistic framework now forms the basis for unraveling how environmental stresses trigger deviations (e.g. false rings) from the normal tree-ring structure.

Introduction

Wood performs three essential functions in trees: it supports and spatially distributes photosynthetic tissues above ground (Fournier *et al.*, 2006); it conducts water and nutrients from the roots to the leaves (Sperry *et al.*, 2008); and it stores carbohydrates, water and defensive compounds (Kozlowski & Pallardy, 1997). Genetic evolution balances these various functions on multi-generational to geological time-scales (Carlquist, 1975), and the phenotypic plasticity of wood structure results in marked adjustments over an individual lifespan or even from year to year (Chave *et al.*, 2009; Fonti *et al.*, 2010). Trade-offs in these various functions, or limits of 'multitasking', are readily apparent. In conifers, for example, > 90% of the wood is composed of tracheid cells, which must simultaneously provide water transport and mechanical support (Brown *et al.*, 1949). Large-diameter tracheids are more efficient at transporting water, but are also more prone to cavitation (Pratt *et al.*, 2007); thick-walled tracheids provide strong mechanical support, but are less capable of transporting water (Cochard *et al.*, 2004; Sperry *et al.*, 2006). Balancing these structural and physiological trade-offs has resulted in a typical tree-ring structure, which can be found in all conifer species under mild environmental conditions (Schweingruber, 1990; Schoch *et al.*, 2004). The conifer tree-ring structure is characterized by a transition from wide, thin-walled earlywood cells produced at the beginning of the growing season, to narrow, thick-walled latewood cells formed at the end of the growing season (Fig. 1). These changes in tracheid morphology drive many fundamental structural and functional tree-ring properties, including density (Rathgeber *et al.*, 2006).

Xylogenesis is the key process during which trees balance the aforesaid functional and structural trade-offs, and fix them permanently in the wood tissue. In recent years, much effort has been given to the molecular biology, biochemistry and biophysics of xylogenesis. Such advances include the description of the biosynthetic pathways of cellulose (Somerville, 2006; Mutwil *et al.*, 2008) and lignin (Boerjan *et al.*, 2003; Vanholme *et al.*, 2010), identification of the molecular regulatory mechanisms at the genetic (Du & Groover, 2010; Zhong *et al.*, 2011) or hormonal (Uggla *et al.*, 1996; Nilsson *et al.*, 2008) levels, and the study of the cellular ultrastructure (Chaffey *et al.*, 2002; Oda & Hasezawa, 2006). It is thus striking how the details of xylogenesis are being intensively studied and increasingly understood, whereas general aspects of wood formation dynamics remain insufficiently known to yield an integrated understanding of how wood is formed (Vaganov *et al.*, 2011).

In his pioneering work, Skene (1969) set the framework for studying the kinetics of tracheid development: the tracheid's final radial diameter is the result of the duration and rate of cell enlargement, whereas the amount of secondary cell wall is a function of the duration and rate of wall material deposition. Subsequent studies, however, emphasized the duration of the processes (cell

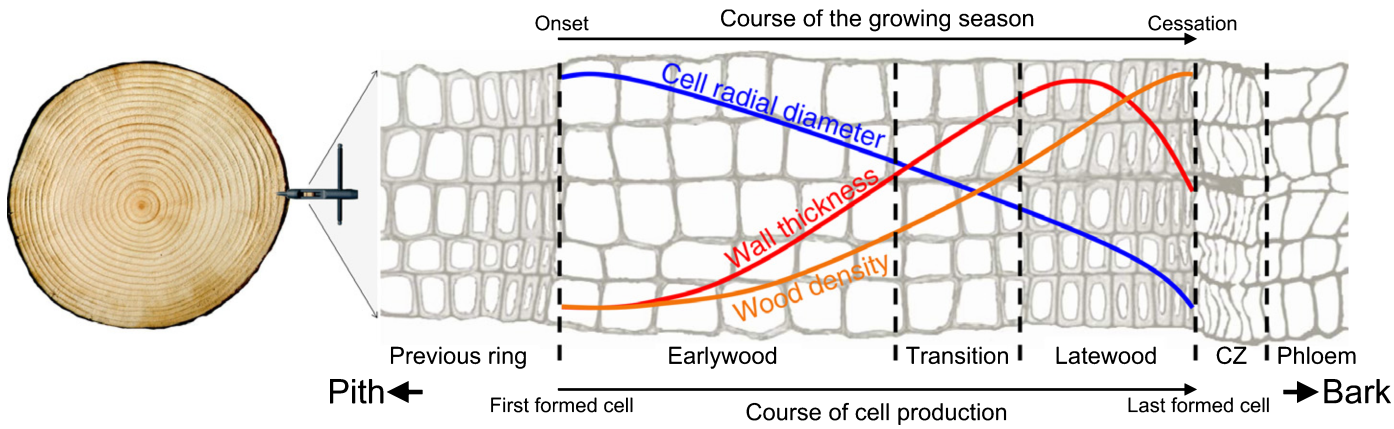

Fig. 1 Typical tree-ring structure generally observed in conifers. Between the latewood and the phloem is the cambial zone (CZ).

enlargement and wall deposition) to explain the observed changes in cell features (cell diameter and wall thickness) along the ring, discounting the importance of rates without assessing them (Wodzicki, 1971; Denne, 1972; Skene, 1972; Dodd & Fox, 1990; Horacek *et al.*, 1999). Consequently, these studies failed to unravel how the duration and rate of xylogenesis subprocesses interact to create the observed tree-ring structures.

Because of this lack of integrated understanding, even basic xylogenesis mechanisms have not been consistently considered in environmental (Briffa *et al.*, 1998; Bouriaud *et al.*, 2005; Gagen *et al.*, 2006), ecophysiological (Helle & Schleser, 2004; Offermann *et al.*, 2011) or modeling (Deleuze & Houllier, 1998; Fritts *et al.*, 1999; Ogée *et al.*, 2009; Drew *et al.*, 2010; Eglin *et al.*, 2010) tree-ring-based studies. In contradiction with observations, most of these studies assume that the rates of the xylogenesis processes are the drivers of the system, whereas the durations are fixed or derived directly from the rates. A comprehensive investigation of the relative importance of the rates and durations of xylogenesis subprocesses thus appears a timely and necessary endeavor.

Here, we aim to draw a detailed picture of how the intra-annual dynamics of xylogenesis generates the typical tree-ring structure generally observed in conifers. To reach this goal, we assess the kinetics (duration and rate) of tracheid differentiation processes (enlargement and secondary wall formation) over several growing seasons, and model the relationships between the kinetics and the resulting tracheid features (cell diameter and cell wall thickness) and wood structure (wood density profile). We take advantage of a novel statistical approach (Cuny *et al.*, 2013) to analyze an extensive dataset of 45 trees grown in three mixed stands in the Vosges Mountains (France), monitored over 3 yr (2007–2009) and belonging to three conifer species: Norway spruce (*Picea abies*), Scots pine (*Pinus sylvestris*) and silver fir (*Abies alba*).

Materials and Methods

Study sites, sampling for xylogenesis and slide preparation

Our study was performed on three comparable mixed forests composed of silver fir (*Abies alba* Mill.), Norway spruce (*Picea abies* (L.) Karst.) and Scots pine (*Pinus sylvestris* L.) located between 350 and 650 m above sea level (a.s.l.) in the Vosges Mountains (northeast France). To monitor wood formation, we selected five mature, dominant, healthy trees of each species in each plot for a total of 45 trees (5 trees × 3 species × 3 sites; Supporting Information Table S1).

Tree-ring formation was monitored in 2007, 2008 and 2009 from April to November. Weekly stem microcores were collected from each tree using a trephor (Rossi *et al.*, 2006) and prepared in the laboratory for histological observations. Transverse sections (5–10 μm thick) were cut with a rotary microtome (HM 355S; MM France, Francheville, France), stained with cresyl violet acetate and permanently mounted on glass slides using Histolaque LMR® (for details on sampling strategy and microcore preparation, see Cuny *et al.*, 2012).

Histological observations

Approximately 4300 anatomical sections were observed using an optical microscope (AxioImager.M2; Carl Zeiss SAS, Marly-le-Roi, Marly-le-Roi, France) at × 100–400 magnification to reconstruct the temporal succession of tracheid differentiation over the growing season. Cells in the cambial, enlargement, wall-thickening and mature zones were differentiated and counted along at least three radial files. Thin-walled enlarging cells were discriminated from cambial cells by their larger size. Moreover, under polarized light, they presented dark primary walls, unlike wall-thickening cells, which exhibited birefringent secondary walls (Abe *et al.*, 1997). Walls of thickening cells appeared violet and blue, indicating that lignification was in progress, whereas entirely lignified walls of mature cells were completely blue.

According to Rossi *et al.* (2003), count data were standardized by the total number of cells of the previous ring using the R package CAVIAR (Rathgeber, 2012). This standardization process reduced the noise in the data, increasing the signal-to-noise ratio by nearly 50% on average (Table S2).

Tracheid anatomy and calculation of morphometric density

For each tree, a digital image of a well-preserved section of the entire ring was recorded at the end of the growing season. It was

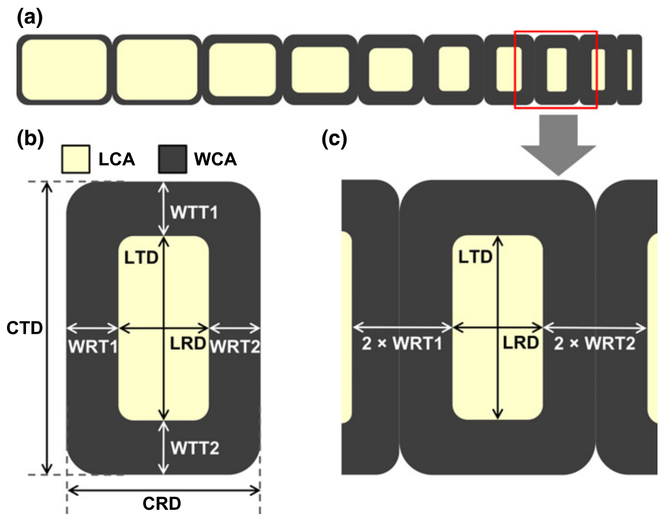

Fig. 2 Presentation of tracheid dimensions and their measurement. (a) A theoretical radial file of tracheids along a conifer ring. (b) The different dimensions of a tracheid in a transversal plan, with the cell radial (CRD) and tangential (CTD) diameters, the lumen radial (LRD) and tangential (LTD) diameters, the double wall radial (WRT1 and WRT2) and tangential (WTT1 and WTT2) thicknesses, and the lumen (LCA) and wall (WCA) cross-sectional areas. The cell cross-sectional area is not represented. (c) The measurement made by the WinCell software, which, for each tracheid along the radial file, measures LRD, LTD and the distances between the lumens of the neighboring tracheids, from which WRT1 and WRT2 are obtained. WinCell also measures the lumen cross-sectional areas.

Table 1 List of the variables used in this work

Notation	Variable	Unit	Acquisition
LRD	Lumen radial diameter	µm	Measured
LTD	Lumen tangential diameter	µm	Measured
LCA	Lumen cross-sectional area	μm^2	Measured
WRT	Wall radial thickness	µm	Measured
WTT	Wall tangential thickness	µm	WTT = 1.2 × WRT
WCA	Wall cross-sectional area	μm^2	WCA = CCA − LCA
CRD	Cell radial diameter	µm	CRD = LRD + 2 × WRT
CTD	Cell tangential diameter	µm	CTD = LTD + 2.4 × WRT
CCA	Cell cross-sectional area	μm^2	CCA = CRD × CTD
MC	Mork's criterion	Unitless	$MC = \frac{4 \times WRT}{LRD}$
MD	Morphometric density	$kg\ m^{-3}$	$MD = \frac{WCA}{CRD \times CTD} \times 1000$
XD	X-ray measured density	$kg\ m^{-3}$	Measured
d_E	Duration of cell radial enlargement	d	Calculated
d_W	Duration of wall thickening	d	Calculated
r_E	Rate of cell radial enlargement	$\mu m\ d^{-1}$	$r_E = \frac{CRD}{d_E}$
r_W	Rate of wall deposition	$\mu m^2\ d^{-1}$	$r_W = \frac{WCA}{d_W}$

used to measure: (1) the lumen radial diameter; (2) the wall radial thickness; (3) the lumen cross-sectional area; and (4) the lumen tangential diameter of all successive tracheids along at least three radial files (Fig. 2). Measurements were performed using WinCell (Regent Instruments, Québec, Canada). During the enlargement phase, cell size increases primarily in the radial direction, whereas the tangential diameter and length remain almost constant (Skene, 1972). Therefore, we used the cell radial diameter (CRD) as a measure of the final outcome of the enlargement phase. CRD was calculated as the sum of the lumen radial diameter and the double radial wall thickness (Table 1, Fig. 2). In addition, the cell tangential diameter and cell cross-sectional area were estimated based on a constant ratio of 1.2 between the tangential and radial wall thickness (Skene, 1972; Rathgeber *et al.*, 2006) and assuming rectangular-shaped tracheids (Table 1, see Methods S1 for a detailed explanation). By contrast, the final wall thickness depends on the amount of wall material deposited and the volume of the cell to be covered (Skene, 1969). Moreover, although cell volume is constant after enlargement, the deposition of material on the inner side of the cell wall reduces the volume of the lumen and hence the surface of deposition. Therefore, instead of the more commonly applied wall thickness metric, we used the wall cross-sectional area (WCA) – a direct estimate of the amount of deposited material per cell – as the final outcome of the wall-thickening phase.

To show variation in tracheid dimensions along a ring, cell morphological measurements were grouped by radial file in profiles called tracheidograms (Vaganov, 1990). However, because the number of cells varied among radial files within and between trees, tracheidograms were standardized according to Vaganov's method (Vaganov, 1990) using a dedicated function of the R package CAVIAR (Rathgeber, 2012). This standardization allows the adjustment of the length of the profiles (cell numbers) without changing their shape (cell dimensions; Vaganov, 1990). We checked visually that this standardization did not alter the shape of the anatomical profiles. The standardized tracheidograms were then averaged for the three plots and over the 3 yr to obtain species-specific tree-ring structures.

Mature tracheids were classified into the different types of wood (earlywood, transition wood and latewood) classically used to obtain easy readable and synthetic descriptions of tree-ring structure. The classification was performed according to Mork's criterion (MC; Denne, 1988), which is computed as the ratio between four times the wall radial thickness and the lumen radial diameter (see the formula in Table 1). Cells were classified as follows: $MC \leq 0.5$, earlywood; $0.5 < MC < 1$, transition wood; $MC \geq 1$, latewood (Park & Spiecker, 2005).

To explore the relationships between cell wall thickness, cell size and the amount of wall material deposited within the cell, we developed a geometric model of tracheid dimensions expressing wall radial thickness as a function of WCA, CRD and cell tangential diameter (see mathematical development in Methods S1).

The influence of tracheid dimension variability on wood density variation along the ring was assessed using the geometric model of Rathgeber *et al.* (2006), which allowed the calculation of the morphometric density from WCA, CRD and the cell tangential diameter (Methods S2).

Wood density measurements

At the end of 2009, two standard cores were taken from each tree, one from each side of the stem (at breast height and perpendicular to the slope), using a Pressler increment borer. A radial strip was cut from each core and exposed to X-ray radiation (Polge & Nicholls, 1972) to obtain a digital map of wood density and tree-ring boundaries using CERD software (Mothe *et al.*, 1998). Each annual ring was divided radially into 20 zones and the mean density of each zone was calculated. Thus, from each ring of each core, we obtained 20 values that described density variation along the ring (i.e. a 'microdensity' profile). The microdensity data of the two cores for each tree were averaged to obtain individual series. The individual microdensity profiles were then averaged for the three stands and years to obtain species-specific microdensity profiles.

Modeling dynamics of wood formation

A very recent methodological breakthrough – based on generalized additive models (GAMs) – allowed us to model the intrinsic complexity and variability of wood formation dynamics (Cuny *et al.*, 2013). Here, we took advantage of this innovative method in order to assess properly the durations and rates of tracheid development. GAMs were fitted to the standardized number of cells in the cambial, enlargement, wall-thickening and mature zones for each year on each individual tree, using the R mgcv package (Wood, 2006). The values of the fitted models were averaged for the three monitored sites over the three monitored years to calculate means representing the species-specific patterns of wood formation dynamics. For each species, we used the average cell numbers predicted by the GAMs to calculate the date of entrance of each cell into each development zone. From these dates, the residence time of each cell i in the enlargement ($d_{E,i}$) and wall-thickening ($d_{W,i}$) zones was computed (for more details on the computation method, see Cuny *et al.*, 2013). The mean rates of CRD enlargement ($r_{E,i}$) and wall deposition ($r_{W,i}$) were computed for each cell i by dividing its final dimensions by the duration it spent in the corresponding phase (Table 1).

To explore the relationship between tracheid development kinetics and the resulting cell features, we expressed the tracheid dimensions (CRD and WCA) as:

$$\mathrm{CRD} = d_E \times r_E \qquad \text{Eqn 1}$$

$$\mathrm{WCA} = d_W \times r_W \qquad \text{Eqn 2}$$

where d_E and d_W are the durations and r_E and r_W the rates of cell enlargement and wall deposition, respectively.

Sensitivity analyses

Sensitivity analyses were conducted in R (R Development Core Team, 2014) to estimate the relative contributions of each input variable to the results of the different mechanistic models (models of CRD enlargement and wall deposition, and geometric models of wall thickness and wood density). These analyses measured differences in model output when values of input variables were changed one at a time within a range of twice their standard deviations whilst holding the other variables constant at their mean values (Cariboni *et al.*, 2007). Advantages of sensitivity analyses are the possibility to deal with the non-linearity and non-independence of the variables of the model (Saltelli *et al.*, 2004). Contrary to the correlation analyses typically used in previous studies, the results of sensitivity analyses thus allowed us to accurately quantify the individual contributions of the duration and rate of cell enlargement and wall deposition to cell feature and tree-ring structure (CRD, WCA, wall thickness and wood density). To demonstrate the influence of each kinetic parameter on the whole tree-ring structure, sensitivity analyses were illustrated by drawing simulated tree rings. For this, each kinetic parameter was made to vary within its natural range of variations whilst holding the other parameters constant at their mean values. The simulated cell features were used to build virtual radial files assuming rectangular-shaped tracheids. In addition to sensitivity analyses, covariance analyses were performed to test the effect of species on the relationships between variables.

Results

Tree-ring structure

All three studied species (silver fir, Norway spruce and Scots pine) displayed the same characteristic tree-ring structure, with a transition from large, thin-walled cells of light earlywood to narrow, thick-walled cells of dense latewood. Earlywood represented *c.* 40% of the tree rings (Fig. 3), but the transition to latewood was more abrupt in pine: the transition wood represented only 8% of pine tree rings compared with 20% in fir and spruce. A regular decrease in CRD was observed from the beginning to the end of the rings (Fig. 3a); the first earlywood cells were three times larger (*c.* 45 μm) than the last latewood cells (*c.* 15 μm). Because the tangential diameter was almost constant (35 ± 2 μm, mean ± SD), the changes in cell cross-sectional area followed those of CRD (Figs 3b, S1). The tracheid wall thickness was lowest (2.7 ± 0.2 μm) in the first 25% of the rings, increased to a maximum of 6 μm in latewood, and decreased abruptly to a minimum of *c.* 4 μm in the last cells of the rings (Fig. 3c). By contrast, WCA increased slightly in the first 75% of the rings, and then decreased abruptly to its minimum in the last latewood cells (Fig. 3d). Sensitivity analysis of the geometric model of cell wall thickness (Methods S1) indicated that 67% of the changes in wall thickness along a ring could be attributed to changes in CRD, and only 33% to changes in WCA (Table S3). These anatomical properties were broadly reflected in the integrative X-ray density (Table S4, Fig. S1). Density profiles (Fig. 3e) were S-shaped, with

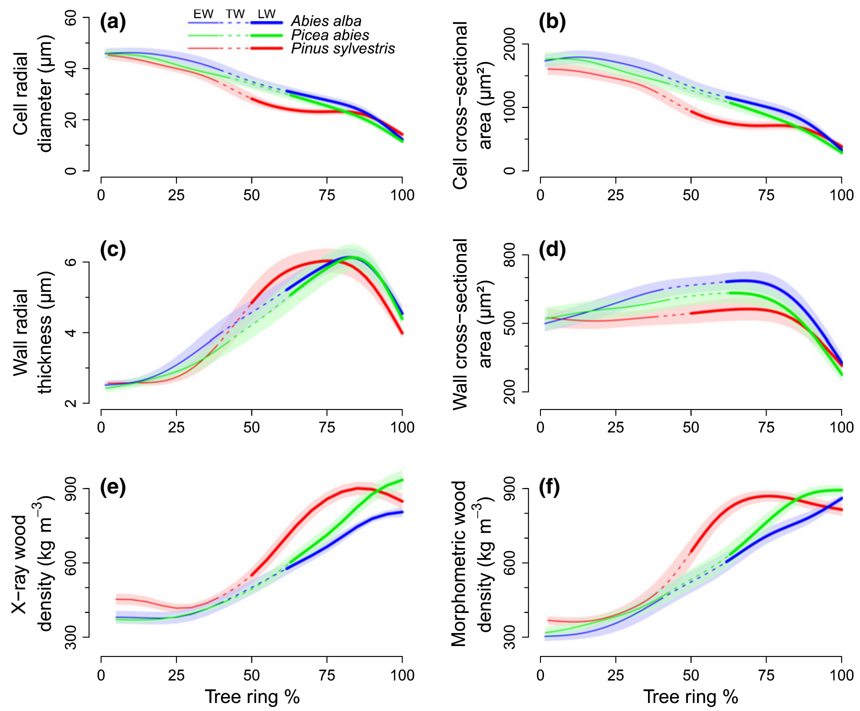

Fig. 3 Tree-ring structure of the three conifer species (*Abies alba*, *Picea abies* and *Pinus sylvestris*), with the variations in cell radial diameter (a), cell cross-sectional area (b), wall radial thickness (c), wall cross-sectional area (d), wood density measured by X-ray (e) and morphometric wood density (f) along tree rings divided into earlywood (EW), transition wood (TW) and latewood (LW). For each species, the line represents the mean values for 15 trees over 3 yr (2007–2009), and the shadowed area delimits the 90% confidence interval.

minimal values in earlywood (*c.* 375 kg m^{-3} in fir and spruce and 450 kg m^{-3} in pine) and maximum values in latewood (800, 900 and 930 kg m^{-3} in fir, pine and spruce, respectively). Only in pine was the density decreased slightly in the last 15% of a ring. The morphometric density computed from the tracheid dimensions was in agreement with the X-ray density (Figs 3f, S1). Sensitivity analysis of the geometric model of wood morphometric density (Methods S2) suggested that changes in CRD accounted for 75% of the changes in wood density along the rings, whereas variation in WCA accounted for the remaining 25% (Table S5).

Wood formation dynamics

Wood formation dynamics captured using GAMs revealed distinctive patterns for the seasonal changes in cell number in the different developmental zones. The number of cells counted weekly in the cambial and enlargement zones both followed bell-shaped curves skewed to the left (Fig. 4a,b). The number of cambial and enlarging cells rapidly increased from April onwards to culminate in May, after which it decreased slowly until mid-September. By contrast, the number of cells in the thickening phase followed a bimodal, right-skewed curve (Fig. 4c). A rapid increase in spring led to a first peak at the beginning of June. A second increase in summer resulted in a maximum in August. From then on, the number of thickening cells decreased rapidly until November. The number of mature cells followed two periods of rapid accumulation separated by a period of slower accumulation (Fig. 4d). The first mature cells were completed during the second half of May. Earlywood cells reached maturity during the initial period of fast accumulation, from spring until mid-July; transition wood cells matured during the intermediate period of slower increase (mid-July to mid-August); finally, latewood cells were completed during the last period of steady accumulation between late August and late October.

Kinetics of tracheid development

The basic observations on the seasonal changes in cell number in the different zones of the maturing tree ring presented above are crucial to the quantification of the kinetics of cell enlargement and wall deposition processes. Differentiating xylem cells needed, on average, 9 d at a rate of 4.2 µm d^{-1} to fully enlarge (Fig. 5). Yet, the duration of cell enlargement (d_E) in pine was 40% longer than in fir and spruce, and was associated with a 40% lower enlargement rate (r_E). Despite these differences in levels, d_E followed similar variations along the ring for the three studied species: it decreased progressively by *c.* two-thirds from the first to the last cells of the rings (from 18 to 6 d in pine, 14 to 5 d in fir and 12 to 4 d in spruce; Fig. 5a). The decrease was particularly abrupt for transition wood cells in pine. By contrast, r_E showed little intra-seasonal variation, except in fir (Fig. 5b).

On average, secondary wall formation required slightly > 1 month, with very small variations between species (d_W = 35 ± 13 d). The average rate of wall material deposition (r_W) was *c.* 19 ± 7 µm^2 d^{-1}. The duration of secondary wall formation increased from 20 to 55 d along the successively produced cells, exhibiting the same stretched S-shaped curve for the three species (Fig. 5c), whereas r_W exhibited the opposite trend (Fig. 5d). Secondary wall formation of earlywood, transition wood and latewood tracheids required *c.* 22 ± 3, 29 ± 4 and 50 ± 6 d, respectively. In earlywood cells, r_W remained stable at 25 ± 2 µm^2 d^{-1}, and then decreased continuously during the production of transition wood and latewood cells to a minimum of *c.* 5 µm^2 d^{-1} for the very last cells.

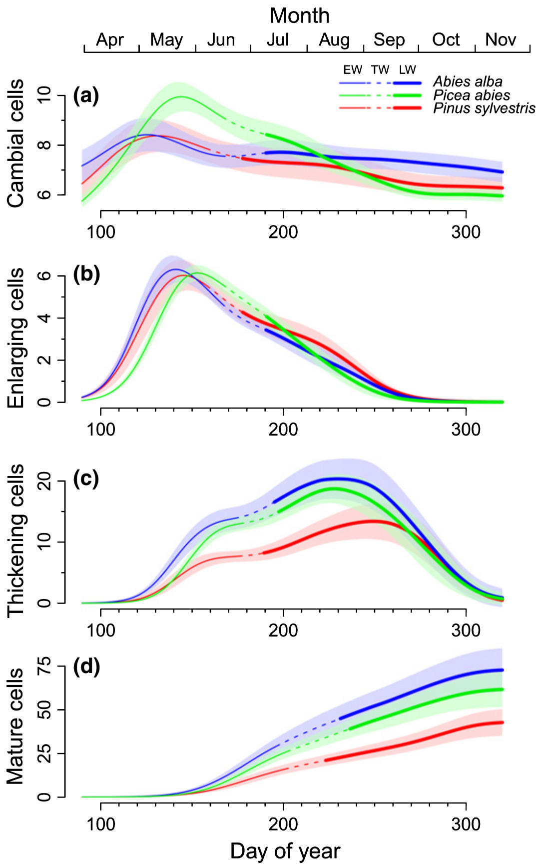

Fig. 4 Generalized additive models (GAMs) fitted to the number of cells counted weekly in the cambial (a), enlargement (b), wall-thickening (c) and mature (d) zones for *Abies alba*, *Picea abies* and *Pinus sylvestris*. For each species, the line represents the mean values for 15 trees over 3 yr (2007–2009), and the shadowed area delimits the 90% confidence interval. The different line types represent the separation between earlywood (EW), transition wood (TW) and latewood (LW) cells.

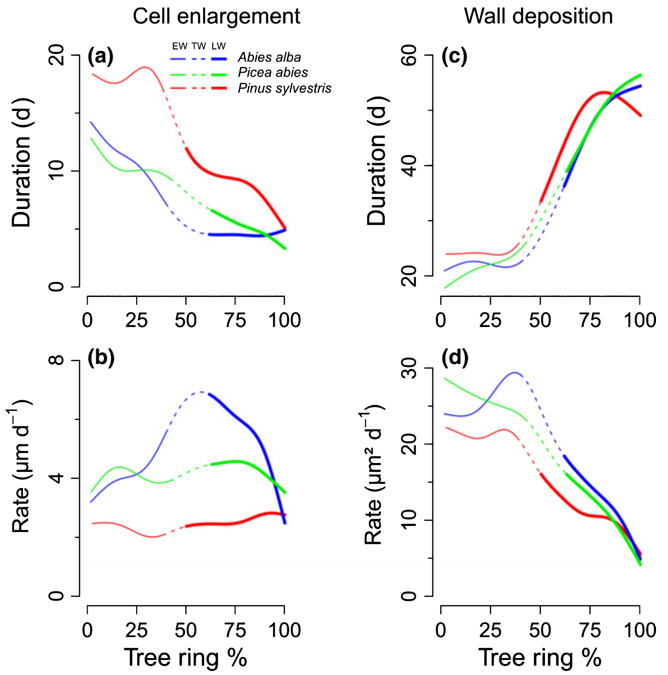

Fig. 5 Kinetics of cell development for the successive tracheids along tree rings of *Abies alba*, *Picea abies* and *Pinus sylvestris*, with the duration of cell enlargement (a), rate of cell enlargement (b), duration of wall deposition (c) and rate of wall deposition (d). The different line types represent the separation between earlywood (EW), transition wood (TW) and latewood (LW) cells.

These data revealed noticeable patterns in the relationships between the duration and rate of cell enlargement and wall deposition. Although d_E and r_E were not correlated (Fig. 6a), a very significant and non-species-specific relationship linked d_W and r_W ($P < 0.001$, $r^2 = 0.92$, $n = 177$; Fig. 6b). This negative relationship held for the majority of the cells of a ring, with the exception of the last few latewood cells. For these cells, the d_W increase reached a plateau, whereas the r_W decrease accelerated (Fig. 6b).

Relationships between the kinetics of tracheid development and tree-ring structure

We found a highly significant, linear and species-specific relationship between CRD and the duration of the enlargement phase (d_E; $P < 0.001$, $r^2 = 0.85$, $n = 177$; Fig. 7a, Table S6): the longer d_E, the larger CRD. By contrast, the rate of cell enlargement (r_E) remained roughly constant whatever the final size of the tracheid (Fig. 7b). Only in fir did the relationship between CRD and d_E or r_E present a hyperbolic shape.

The sensitivity analysis of the model of cell enlargement ($CRD = d_E \times r_E$) allowed us to quantify that the variation in d_E along the rings contributed to 75%, on average, of the changes in CRD, whereas r_E contributed to only 25% (Table S7). The studied species, however, presented species-specific modulations of this general trend. Thus, in agreement with the slightly different behaviors observed for fir in Fig. 7(a,b), we found a lower contribution of d_E and a higher contribution of r_E in fir than in pine and spruce (65%/35%, 78%/22% and 81%/19%, respectively).

A similar analysis of the wall deposition model ($WCA = d_W \times r_W$) suggested that the variation in d_W and r_W along the rings contributed equally to the changes observed in WCA, independent of the species (Table S8). WCA was not correlated with either d_W or r_W (Fig. 7c,d).

The combined results of the sensitivity analyses allowed us to estimate the contribution of the different kinetic parameters to the changes in tracheid dimensions and wood density along the rings (Fig. 8). We found that d_E contributed to 75% of the changes in CRD, which, in turn, contributed to 67% of the changes in wall thickness. Therefore, d_E contributed to 50% (75% of 67%) of the changes in wall thickness along the rings. Similarly, d_E contributed to 56% (75% of 75%) of the changes in wood density. Proceeding in the same way, we found that r_E, d_W and r_W contributed to 19%, 12.5% and 12.5%, respectively, of the changes in wood density along the rings.

Discussion

This study combined an original and extensive dataset (45 trees from three conifer species monitored during 3 yr) with recent methodological advances in modeling techniques to accurately

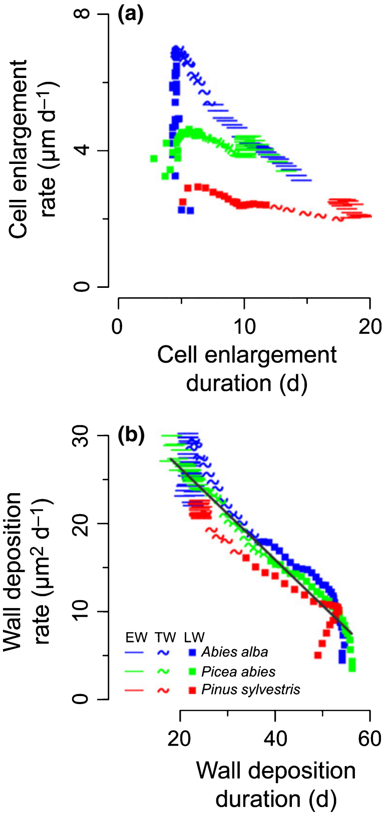

Fig. 6 Relationships between the duration and rate of cell enlargement (a) and the duration and rate of wall deposition (b) in *Abies alba*, *Picea abies* and *Pinus sylvestris*. For each species, each point represents a single cell along an average ring (mean values for 15 trees over 3 yr). The different point types discriminate earlywood (EW), transition wood (TW) and latewood (LW) cells. In (b), the black line represents the non-species-specific linear relationship between the duration and rate of wall deposition.

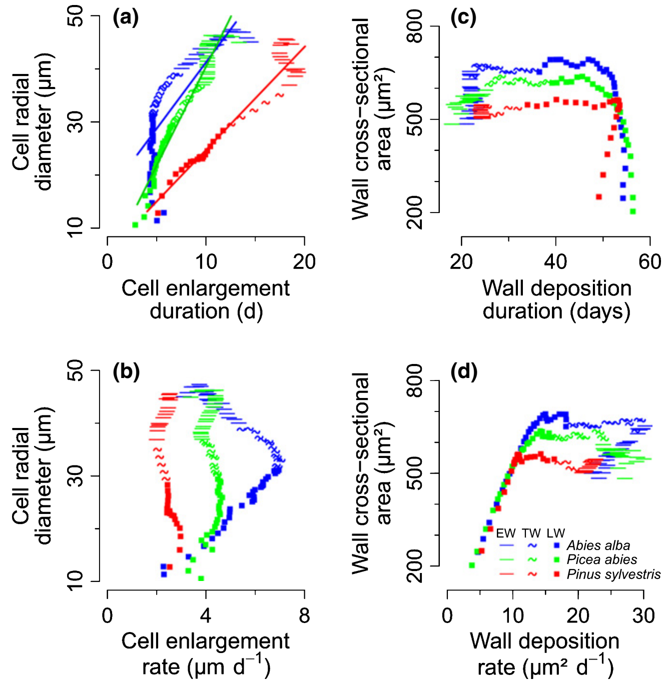

Fig. 7 Relationships between the kinetics of xylogenesis processes and the anatomical features of tracheids in *Abies alba*, *Picea abies* and *Pinus sylvestris*. Relationships between the duration of cell enlargement and the final cell radial diameter (a), the rate of cell enlargement and the final cell radial diameter (b), the duration of wall deposition and the final wall cross-sectional area (c) and the rate of wall deposition and the final wall cross-sectional area (d). For each species, each point represents a single cell along an average ring (mean values for 15 trees over 3 yr). The different point types discriminate earlywood (EW), transition wood (TW) and latewood (LW) cells. In (a), lines represent the species-specific linear relationships between the duration of enlargement and the cell radial diameter.

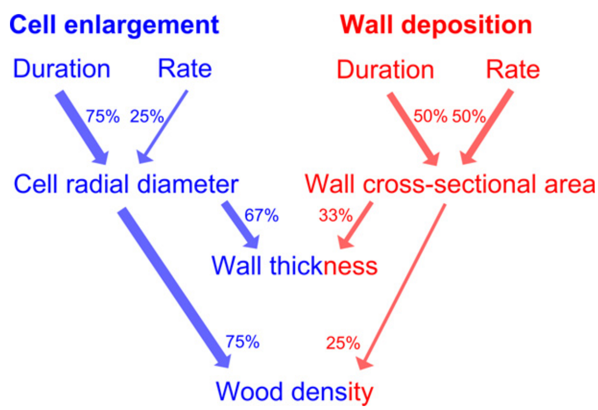

Fig. 8 Relative contributions of the different components of the kinetics of tracheid development to the changes in tracheid dimensions and wood density along the ring. The percentages provided were calculated on the basis of sensitivity analyses.

investigate the intra-annual dynamics of wood formation. This work allowed us to quantify concomitantly the contribution of both the durations and rates of xylogenesis subprocesses to the anatomical changes observed along the rings, and thus to draw the first detailed picture of how the kinetics of tracheid development creates the typical tree-ring structure that characterizes conifers grown in the temperate zone when no particular stresses are involved (Fig. 8).

Rethinking tree-ring formation in conifers

One major result obtained in this study questions the relationship between cell wall thickness and wood density. These two parameters are generally considered to reflect the amount of biomass allocated to xylem cells. For example, the dense latewood with thick-walled cells is thought to hold most of tree biomass (Begum *et al.*, 2012), and several authors have attempted to associate latewood formation with carbohydrate availability (Larson, 1964; Uggla *et al.*, 2001; Rossi *et al.*, 2009). Therefore, patterns of wall thickness and wood density along conifer rings are regularly assumed to be primarily driven by the process responsible for filling the cells with biomass, namely wall deposition. Based on this concept, wall thickness changes generally have been related to the kinetics of wall deposition (Wodzicki, 1971; Denne, 1972; Dodd & Fox, 1990; Horacek *et al.*, 1999), with the idea that the wall thickness is mainly driven by the wall-thickening duration anchored in the scientific literature (e.g. Uggla *et al.*, 2001; Bouriaud *et al.*, 2005; Vaganov *et al.*, 2011). In opposition to this common belief, our results revealed that the amount of material deposited per cell was almost constant along most of the ring, but decreased dramatically in the last 25% of the ring to reach minimal values for the last latewood cells, where wood density was maximal. Because of this, changes in the amount of wall material per cell, and in the kinetics of wall deposition, explained only 33% of the changes in wall thickness and only 25% of the changes in wood density. This result forces us to completely rethink the way in which xylogenesis creates tree-ring structure in conifers.

In particular, our results emphasized the crucial role of the decrease in cell size in driving changes in wall thickness and wood density. Cell size contributed to 67% of the increase in wall thickness and to 75% of the increase in wood density along the ring. These results can be explained by the geometric effect of cell size diminution. Namely, more or less constant amounts of wall material are placed into smaller and smaller lumen volumes, resulting in increasing wall thickness, cell to wall ratio and wood density. Therefore, counter-intuitively, along a conifer ring, increasing cell wall thickness and wood density do not reflect higher wood biomass allocation; they mostly occur because the volumes in which the biomass is allocated decrease. Therefore, the main driver of wall thickness and density changes along a ring is the cell enlargement process.

Most experimental studies on the kinetics of tracheid differentiation show that the typical decrease in cell size along the tree ring parallels an overall decrease in the duration of cell enlargement (Wodzicki, 1971; Skene, 1972; Horacek *et al*., 1999), whereas the rate of cell enlargement presents fewer variations and more irregular patterns (Wodzicki, 1971; Skene, 1972). By providing an unambiguous assessment of the relative contributions of the duration and rate of enlargement to changes in cell size, our results now quantitatively establish the duration of enlargement as the main driver of cell size decrease. The changes in the duration of enlargement explained 75% of the changes in cell size along the ring, whereas the changes in the rate explained the remaining 25%. It is interesting to note that several authors have also related the tapering of tracheid diameter along the stem to changes in the duration of enlargement (Denne, 1972; Dodd & Fox, 1990; Anfodillo *et al*., 2012).

To visually demonstrate the role of the duration and rate of cell enlargement and wall deposition on tree-ring structure, we used the results of previous sensitivity analyses to simulate the formation of virtual radial files (Fig. 9). Only the change in the duration of enlargement over the growing season is able to create 'realistic' virtual rings whose structure is like those typically observed for natural conifer tree rings (Fig. 9a,b). By contrast, the other kinetic parameters lead to unrealistic tree-ring structures (Fig. 9c–e). A change in the rate of enlargement, with other parameters fixed, produces mainly transition wood; a change in the duration of wall deposition reproduces the wall thickness increase, but cells have a constant size; a change in the rate of cell wall deposition inverts earlywood and latewood.

The same line of reasoning can explain variations in wood structure among the species. Scots pine, in comparison with Norway spruce and silver fir, exhibits a sharper transition

Fig. 9 Schemes of the different tree rings formed when the different parameters of the kinetics of tracheid development are changed one at a time.

between earlywood and latewood (Schweingruber, 1990; Schoch *et al*., 2004). This species-specific anatomical feature also originates in enlargement duration particularities. Indeed, in pine, a prominent decline in the enlargement duration characterized the cells of transition wood, whereas the decrease was more gradual in fir and spruce. Thus, using only the duration of the enlargement phase, we were able to simulate the species-specific transitions for pine and spruce (Fig. 9b). In fir, the duration of enlargement was not sufficient to simulate the gradual transition, which can be explained by the greater importance of the enlargement rate in this species.

The kinetic model presented here has been established on three major European conifer species, whose typical tree-ring structure is representative of most conifers (Schweingruber, 1990; Schoch *et al*., 2004). We believe that the conceptual framework provided by this model can be applied to many species and other environments. However, further investigations are needed to understand how the relationships between cell development kinetics and wood anatomy are modified for conifer species with narrow latewood (e.g. *Pinus cembra* or *Juniperus* species) or rings formed under stressful conditions (e.g. xeric site, drought year, insect attacks, late frosts) and typically associated with wood anatomical anomalies (e.g. light latewood, false rings).

New insights into the regulation of xylogenesis

Another important outcome of this study is the quantification of the relationships between the durations and rates of processes, which provides new insights into the regulation of xylogenesis. We found that the duration and rate of cell enlargement were not correlated (Fig. 6a), which was not expected. It is intuitive that the duration and rate of enlargement should be coupled because they are controlled by the same phytohormones (e.g. auxins, cytokinin, gibberellin). For example, auxin is distributed along a steep concentration gradient across the division and enlargement zones and acts as a morphogen (Uggla *et al*., 1996). The width of the gradient determines the width of the enlargement zone, and thus controls the duration of the cells in the zone (Tuominen *et al*., 1997; Sundberg *et al*., 2000). Auxin also stimulates cell enlargement by inducing extracellular acidification, which activates expansins, proteins that loosen the cell wall to aid enlargement (Cosgrove, 2000; Perrot-Rechenmann, 2010). However, the fact that the same set of agents controls both the duration and rate of a subprocess does not necessarily lead to a coupling of kinetics, as demonstrated by the absence of relationships in our data. Increasing the level of auxin, for example, should increase the width of the enlargement zone, but, because of the stimulating effect of auxin on cell growth (in the enlargement and division zones), it should also increase the 'speed' at which cells are 'crossing' this zone. Such a complex and dynamic system highlights the need for detailed models patterning plant cell development (including xylogenesis) to take into account the intertwined effects of regulatory agents in the different developmental phases (Harashima & Schnittger, 2010).

In contrast with cell enlargement, the duration and rate of secondary cell wall deposition exhibited a strong negative correlation (Fig. 6b). The decrease in the rate of deposition during the growing season was counterbalanced by the increase in the duration. This results in a nearly constant amount of deposited material, except for the last quarter of the ring (Fig. 2d). Groover & Jones (1999) revealed a biological mechanism coordinating secondary cell wall synthesis and apoptosis that provides an explanation of the observed coupling. During the secretion of wall material, a protease accumulates in the extracellular matrix. Once a critical concentration of protease is reached, cell death is triggered and secondary wall formation stops. According to this mechanism, the seasonal decreasing rate of wall deposition observed will be accompanied by a decreased excretion of protease in the extracellular matrix. Thus, as the growing season progresses, more time will be required for protease to reach the critical level, delaying apoptosis and increasing proportionally the duration of wall deposition. For the last latewood cells, however, the coupling between rate and duration is lost. At this stage, the deposition rate may be too low for the concentration of protease to reach the critical threshold. A secondary mechanism may take over to trigger apoptosis, such as sucrose concentration or reactive oxygen species (Bollhöner *et al*., 2012).

Conclusions

We provided novel quantifications and associated mechanistic models of wood formation kinetics that explain the formation of the tree-ring structure normally (i.e. in the absence of major stress) observed in conifers. Importantly, our study refutes the long-standing assumption that the increase in wall thickness and wood density along the ring is driven by the fixation of more biomass during the wall-thickening process. Instead, we demonstrate that the amount of wall material per cell is almost constant along a ring. Consequently, changes in wall thickness and wood density are principally driven by changes in cell size. The duration of cell enlargement contributes to 75% of the changes in cell size, whereas the rate contributes to the other 25%. This mechanistic framework – developed in this study to understand the formation of the typical, widely observed, conifer tree-ring structure – will also provide the basic tools for unraveling how climatic variations, extreme events and, more generally, environmental stresses lead to deviations from the 'normal' tree-ring structure, including variability in the maximum latewood density and the occurrence of false rings.

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Supporting Information

Additional supporting information may be found in the online version of this article.

Fig. S1 Relationships between variables of tree-ring structure.

Table S1 Characteristics of the monitored trees

Table S2 Comparison of the signal-to-noise ratio between raw and standardized xylem cell count data

Table S3 Sensitivity analysis of the geometric model of tracheid dimensions

Table S4 Relationships between tracheid dimensions and wood density

Table S5 Sensitivity analysis of the geometric model of wood density

Table S6 Analysis of covariance of the relationship between the duration of cell enlargement and final radial cell diameter

Table S7 Sensitivity analysis of the model of tracheid enlargement

Table S8 Sensitivity analysis of the model of secondary wall formation

Methods S1 Development of the geometric model of tracheid dimensions.

Methods S2 Development of the geometric model of wood density.

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